A Double-Head Global Reasoning Network for Object Detection of Remote Sensing Images

Jingye Shi, Ruicong Zhi, Jingru Zhao, Yuewu Hou, Zeyong Zhao, Jingyi Wu, and Jiqiang Liu

***Abstract*— Object detection of remote sensing images is a fundamental and important task for a wide range of applications, such as agriculture, military, and geological exploration. However, it is still a challenging task due to complex background, heavy occlusion, and object ambiguities. Complex remote sensing scenes contain rich visual and spatial relationships between proposals, while most existing methods ignore these relationships. In this article, we introduce a novel double-head global reasoning network (DGRN), which endows an object detection model with the ability to adapt global reasoning by propagating visual and spatial embeddings between positive proposals (foreground) and negative proposals (background). Instead of propagating the features of positive proposals, we evolve high-level embeddings globally to explore potential information in all proposals. Specifically, our method first classifies and locates positive proposals and negative proposals simultaneously, and then adaptively learns two sparse region-to-region undirected graphs for classification and location, respectively. Finally, the graph reasoning module (GRM) conducts the propagation of proposals' embedding to improve the performance of object classification and localization. Without any prior knowledge, our DGRN method explores reasoning graphs between proposals for object classification and location, respectively, and then uses graph convolutional neural network (GCN) to propagate information and achieve relational reasoning. Our method is light-weighted and flexible enough to enhance many object detection models, making object detection models the capability of relational reasoning. The experimental results on DOTA and DIOR showed that the proposed method had better detection performance than object detection networks without considering relationships.**

***Index Terms*— Double-head global reasoning network (DGRN), graph reasoning, object detection, remote sensing images.**

I. Introduction

OBJECT detection of remote sensing images [1], [2], [3], [4], [5] aims to classify and locate objects of interest (e.g., planes, ships, and vehicles) in complex remote sensing scenes. Remote sensing images record information about the Earth's surface taken by satellites. There are many occluded and ambiguous objects due to different recording conditions, such as different shooting angles, time, and weather. Therefore, object detection of remote sensing images is still a challenging task.

In recent years, deep learning has been successfully applied in object detection of remote sensing images, such as RCNN frameworks [6], [7], [8], [9], [10], [11], [12], You Only Look Once (YOLO) frameworks [4], [13], [14], [15], [16], and SSD frameworks [17], [18], [19], [20]. These models were easily disturbed by low-quality proposals and complex backgrounds. However, is the background in remote sensing images really useless for object detection? Although the backgrounds of remote sensing images are more complex than natural images, the complex background also brings some useful information.

Usually, there is an overlap between positive proposals (foreground) and negative proposals (background), and negative proposals may contain local or global information in the foreground. We believe that reasonably utilizing the relationship between positive proposals and negative proposals can enhance the proposals' features and improve the performance of the object detection model. As shown in Fig. 1, the objects in Fig. 1(a) are large vehicles and small vehicles, and the objects in Fig. 1(b) are ships and a harbor. Due to the similar shape of vehicles and ships, it is easy to cause false detection. However, if we not only focus on the vehicles and the ships but also their contexts, we can get more obvious differences between vehicles and ships from their background. Vehicles are on the road, while ships are on the water, and there are often harbors nearby. By combining the information contained in the foreground and background, vehicles and ships with similar shapes can be better distinguished.

Since there is a significant relationship between positive proposals and negative proposals in remote sensing images, relational reasoning is considered to improve the performance of the object detection model. Relational reasoning is originally used to deal with the problems of long-tail data distributions, heavy occlusions, and pattern ambiguities. However, the existing relational reasoning methods only consider the contribution of positive proposals. Our work aims to develop a reasoning object detection model, which not only incorporates commonsense knowledge from positive proposals but also propagates information including negative proposals.

The existing object detection methods in remote sensing images are limited by treating each object region separately

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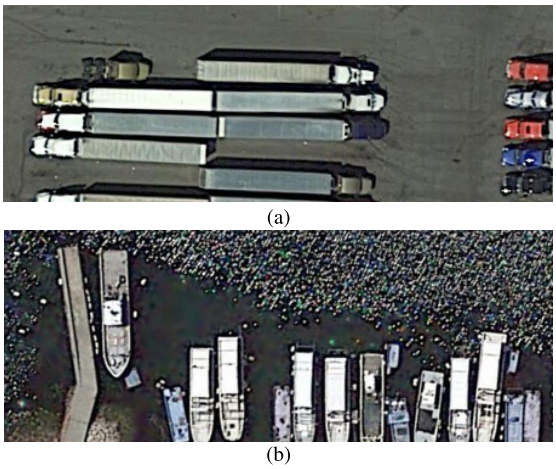


Fig. 1. Different background of (a) vehicles and (b) ships.

without considering crucial semantic dependencies among objects. In this article, we propose a novel double-head global reasoning network (DGRN) that endows object detection modes with the ability of adaptive global reasoning by propagating visual and spatial embeddings between positive proposals (foreground) and negative proposals (background). Instead of propagating only the features of positive proposals, we evolve high-level embeddings globally to explore potential information in all proposals. Without any prior knowledge, our DGRN method explores reasoning graphs between proposals for object classification and location, respectively, and then uses graph convolutional neural network (GCN) to propagate information and achieve relational reasoning. The contributions of our work can be summarized as follows.

1) The contexts of proposals are explored to construct the reasoning graphs to enhance the object detection performance in remote sensing images. On the one hand, it can alleviate the imbalance between positive proposals and negative proposals in remote sensing images. On the other hand, it can effectively explore visual information and spatial information in context.

2) Without adding any prior knowledge, two reasoning graphs between positive proposals and negative proposals are constructed for classification and localization, respectively. The DGRN is light-weighted and flexible to enhance the performance of object detection tasks in remote sensing images. It can be applied to many two-stage or one-stage deep learning-based object detection models to give it the capability of relational reasoning detection.

II. Related Work

Object detection is a fundamental task in computer vision. In the era of deep learning, object detection [21] can be grouped into two genres: region-based (two-stage) frameworks and unified (one-stage) frameworks [22], where the former frames the detection as a "coarse-to-fine" process while the later frames it as to "complete in one step."

A. Generic Object Detection

The two-stage object detection first generates sparse candidate regions on the image, and then extracts features for object classification and bounding-box regression from the candidate regions. Initially, some networks such as MultiBox [23] and RCNN [6] used a classifier to determine the category labels of the proposals based on the features of the proposal boxes extracted by the CNN. Specifically, the classifier used by RCNN was SVM. However, these methods required the CNNS to be run many times after the proposals were extracted, resulting in slow detection. SPPNet [24] sped up inference by adding an SPP layer on top of the last convolutional (CONV) layer to obtain features of fixed length for the FC layers. What's more, Fast RCNN [7] proposed to exchange the order of proposals extraction and feature extraction, and added a regions of interest (ROI) layer to extract a fixed-length feature for each region proposal. The idea of sharing the computation of convolution across region proposals greatly increases the detection speed. Ren et al. [8] proposed the detector Faster R-CNN based on Fast R-CNN. Faster R-CNN introduced a commonly utilized region proposal network (RPN) to take the place of traditional candidate region-generating algorithms to automatically extract better proposals. To further improve the detection speed, Dai et al. [25] proposed a region-based fully convolutional network (R-FCN), which properly allocated the computational expense across the region classification step. Li et al. [26] proposed Light Head R-CNN, making the head of the detection network as light as possible. In order to further increase the accuracy of detection speed. Cai and Vasconcelos [27] proposed Cascade R-CNN based on Faster R-CNN, by cascading multiple detection heads to improve detection accuracy. For more precise positioning, Mask RCNN [28] improved the ROI pooling of Faster R-CNN to ROIAlign layer.

Although two-stage detectors achieved high object detection accuracy, it suffered from the limitation of high computational complexity, which was time-consuming. DetectorNet [29] formulated the detection problem as a bounding box regression problem, but it ran multiple networks for each part on every crop, making it slow. Based on the elegant full convolutional network architecture, the OverFeat [30] was improved in speed. Lin et al. [31] proposed RetinaNet and introduced focal loss. Focal loss enabled the one-stage detectors to achieve comparable accuracy to two-stage detectors while maintaining high detection speed. Similar to DetectorNet, Redmon et al. [13] proposed YOLO, casting object detection as a regression problem. Red et al. proposed YOLOv2, an improved version of YOLO, which replaced the GoogleNet [32] network with DarkNet19 and used K-means to get a good initial anchor. YOLOv3 [14] used a joint training method to increase the number of detected categories and could run in real-time. SSD [17] achieved fast detection speed, while still retaining high detection quality by performing detection over multiple scales CONV feature maps. Novelly, CornerNet [33] formulated bounding box object detection as detecting paired top-left and bottom-right keypoints. YOLO v4 [34] was proposed based on the original YOLO and adopted several optimization strategies that had different degrees of optimization from data processing, backbone, training, activation function, loss function, and other aspects. Ge et al. [35] presented a high-performance anchor-free detector (YOLOX). Equipped with

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a decoupled head, anchor-free, and advanced label-assigning strategy, YOLOX achieved a better tradeoff between speed and accuracy. Wang et al. [36] proposed a unified network (YOLOR) that encode implicit knowledge and explicit knowledge together, and proved that it was still very effective for multitasking learning under the single model architecture. Based on the trainable bag-of-freebies method, Wang et al. [37] developed the YOLOv7 series of object detection systems, which received promising results in object detection. YOLO v8 is the latest addition to Ultralytics' family of YOLO-based object detection models. Leveraging the previous YOLO model, the YOLOv8 model was faster and more accurate while providing a unified framework for performing training models for object detection, instance segmentation, and image classification. In addition, different from CNN-based approaches, DETR [38] proposed a new object detection architecture based on a transformer, which replaced the traditional region proposal-based methods with a fully end-to-end trainable architecture. Deformable DETR [39] proposed a deformable attention module to extract sparse features and accelerate training. Zhang et al. [40] proposed an end-to-end transformer detector (DINO) with contrastive denoising training, mixed query selection, and looked forward twice, which significantly improved both the training efficiency and the detection performance. Yang et al. [41] proposed focal transformers, which demonstrated its superiority in object detection. It presented focal self-attention to enable efficient local-global interactions in vision transformers. The RT-DERT [42] effectively solved the problem of the high computational cost of NMS processing for the DETR model, and established a new end-to-end speed benchmark through an efficient hybrid encoder.

B. Object Detection of Remote Sensing Images

Compared to normally visible images, remote sensing images have a large field of view and high background complexity, and the targets are usually small and rotated with certain angles, which brings great challenges to object detection. Deep learning-based object detection of remote sensing images can also be categorized into rotate box object detection and horizontal box object detection.

The representative methods for horizontal box object detection are as follows. Zheng et al. [43] proposed an adaptive, dynamically refined one-stage detector based on the feature-pyramid transformer, which embedded a transformer in the FPN in order to enhance its feature-fusion capacity. Xu et al. [44] proposed a local-perception backbone based on the Swin transformer for remote sensing images object detection and instance segmentation, and they investigated the performance of their backbone on different detection frameworks. Yang et al. [45] introduced an arbitrary-oriented object detection model and a feature fusion structure suitable for remote sensing images. Yang et al. [46] developed an object detector called RPDet based on RepPoints that achieved competitive object detection performance without the need for anchors. Tian et al. [47] proposed an anchor-free and proposal-free one-stage detector FCOS. By eliminating the predefined set of anchor boxes, FCOS completely avoided the complicated computation related to anchor boxes such as calculating overlapping during training. Xu et al. [48] proposed a simple yet effective framework to detect multioriented objects. Instead of directly regressing the four vertexes, it can accurately describe a multioriented object by gliding the vertex of the horizontal bounding box on each corresponding side. Zhang et al. [49] proposed an adaptive training sample selection (ATSS) to automatically select positive and negative samples according to the statistical characteristics of the object. It significantly improved the performance of anchor-based and anchor-free detectors and bridged the gap between them.

The representative methods for rotating box object detection are as follows. Rotated RPNs (R2PN) [50] generated multi-orientated proposals and predicted the orientation angles of bounding boxes to make the inclined ship region proposals more accurate. Rotation-sensitive regression detector (RRD) [51] proposed to perform classification and regression on features of different characteristics, the regression branch extracts rotation-sensitive features by actively rotating the convolutional filters, while the classification branch extracts rotation-invariant features by pooling the rotation-sensitive features. Song et al. [52] proposed the object-wise point-guided localization detector (OPLD), which introduced a new prediction of center-point to refine the results and avoid the problem of discontinuous regression. Yang and Yan [53] proposed a circular smooth label (CSL) technique to handle the periodicity of the angle and increase the error tolerance to adjacent angles. Yang et al. [54] proposed an end-to-end refined single-stage rotation detector for fast and accurate object detection by using a progressive regression approach from coarse to fine granularity. Han et al. [55] proposed a single-shot alignment network (S^2A-Net) consisting of two modules: a feature alignment module (FAM) and an oriented detection module (ODM). The FAM can generate high-quality anchors with an anchor refinement network. The ODM produced orientation-sensitive and orientation-invariant features to alleviate the inconsistency between classification score and localization accuracy. To solve the problem that remote sensing object detectors require more parameters, resulting in highly redundant and inefficient, Han et al. [56] proposed a rotation-equivariant detector (ReDet), which explicitly encoded rotation equivariance and rotation invariance. The experimental results were better on several challenging aerial image datasets. Xie et al. [57] proposed an effective and simple oriented object detection framework, termed oriented R-CNN, which was a general two-stage oriented detector with promising accuracy and efficiency. AR^2Det [58] adopted two simple modules, ship detector (SDet) and center detector (CDet), to generate and improve the detection results, respectively. Li et al. [59] proposed an effective adaptive points learning approach to aerial object detection by taking advantage of the adaptive points representation, which was able to capture the geometric information of the arbitrary-oriented instances. Experiments showed that the above methods improved the accuracy of the rotating object detection algorithm to a new level.

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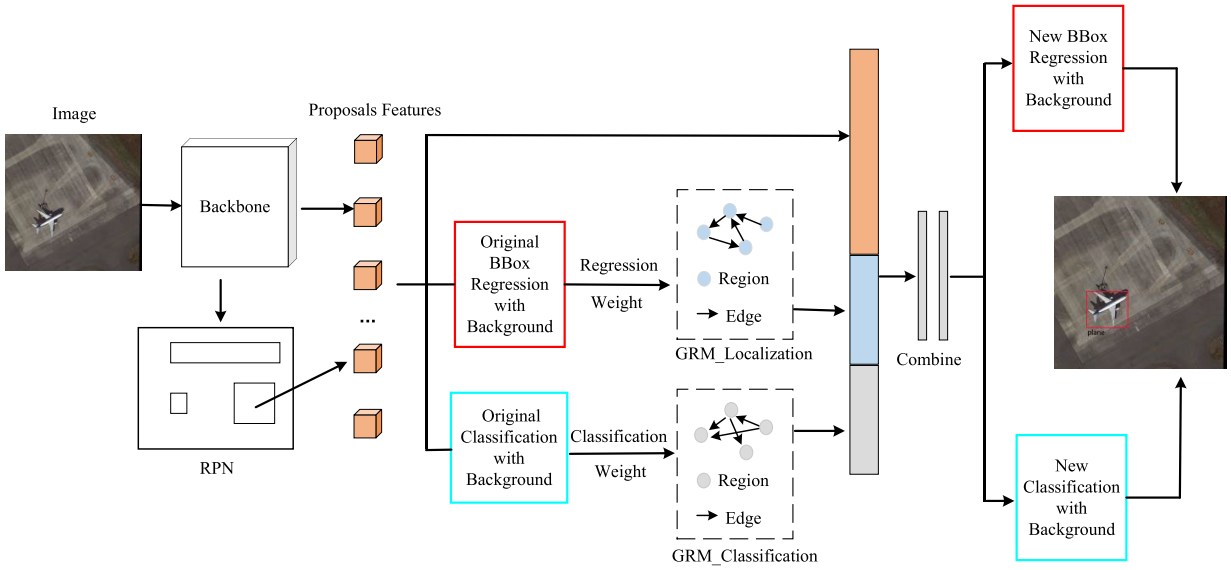


Fig. 2. Overview of the DGRN. DGRN adequately explores the relational information in remote sensing images by detecting reasoning proposals and positive proposals.

III. Proposed Approach

A. Overview

The overall pipeline of DGRN can be found in Fig. 2. First, our method extracts visual features from original remote sensing images by a backbone that is well applied in object detection [8], and a number of proposals are created by the RPN. These proposals are divided into positive proposals, negative proposals, and reasoning proposals according to the intersection-over-union (IoU) of the proposals with ground-truth. Then, positive proposals and reasoning proposals are utilized to exploit the visual information and position information in remote sensing images. The graph reasoning module (GRM) conducts the propagation of proposals' embedding to implement relational reasoning among the proposals. Finally, the output of the GRM is concatenated to the original proposals' features, and fed into multilayer perception to improve the performance of object classification and localization.

B. Foreground and Background Context Learning Module

Up to now, the object detection model of remote sensing images has ignored the background semantic information. Our DGRN discovers background semantic features by detecting reasoning proposals and uses the relationship between foreground and background to improve the representation ability of proposals' features.

1) Reasoning Proposals Detection: In the object detection task, IoU describes the overlap degree of the two boxes. IoU refers to the ratio of the overlapping area and merging area of the two bounding boxes. Generally speaking, when the IoU between a bounding box and a ground-truth is greater than 0.5, the bounding box is regarded as a positive proposal, otherwise, it is assigned a negative label. In the general object detection model, only positive proposals are classified and located, and negative proposals are assigned as background. Due to the complex background of remote sensing images, there is a large imbalance between positive proposals and negative proposals, which reduces the efficiency of the object detection model greatly. In this article, we regard "negative proposals" containing local or global information of ground-truth as reasoning proposals, while "negative proposals" containing a large amount of redundant information are the real negative proposals. We utilize reasoning proposals to capture the useful information contained in the background of remote sensing images, so as to achieve more effective relational reasoning. Fig. 3 showed proposals and ground-truth, Fig. 3(a) showed positive proposals, Fig. 3(b) showed reasoning proposals, and Fig. 3(c) showed negative proposals. Although the IoU proposals are less than 0.5 in Fig. 3(b), they contain local or global information of ground-truth. IoU of the proposals in Fig. 3(c) are almost 0, they contain a lot of redundant information. The threshold is set manually and discussed in Section IV.

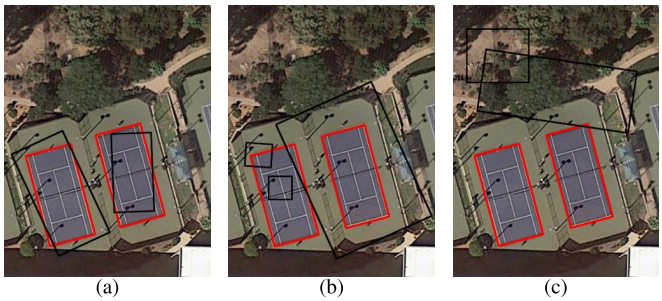


Fig. 3. Different types of proposals. The red box is the ground truth of a tennis court, and the black boxes are the proposals. (a) Positive proposals of tennis court (IoU $\geq$ 0.5), (b) reasoning proposals of tennis court (0.2 $\leq$ IoU $<$ 0.5), and (c) negative proposals of tennis court (IoU $<$ 0.2).

Reasoning proposals for different objects contain different information. In order to fully explore the background information of each class, we construct reasoning classes corresponding to the original object classes.

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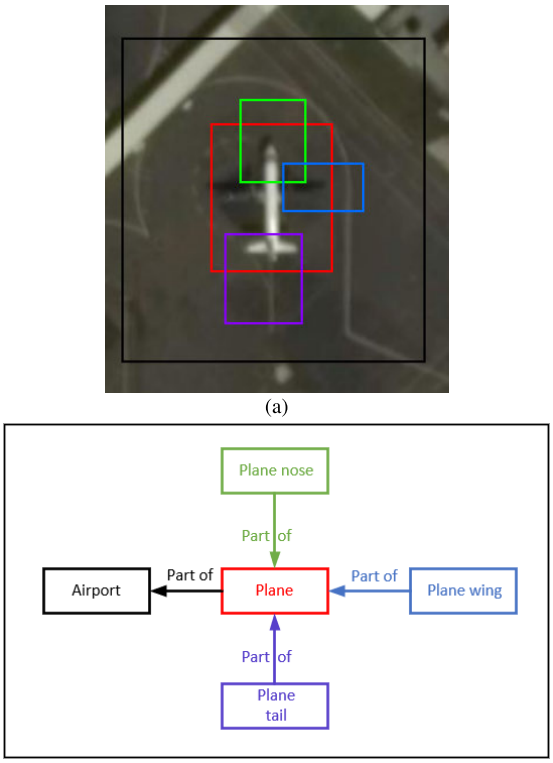
Reasoning proposals can be regarded as "hard samples." There are several benefits to exploring latent content in reasoning proposals, which are ignored in the general object detection model. On one hand, reasoning proposals effectively alleviate the imbalance between positive proposals and negative proposals in remote sensing images. On the other hand, detecting "hard samples" could improve the robustness of the object detection model.

2) Double-Head Reasoning Strategy: Object detection consists of two subtasks: classification and localization. Classification focuses on the visual information, while localization focuses on the spatial information. Therefore, a double-head reasoning strategy is introduced to deal with the two subtasks, respectively. In short, our method utilizes visual and spatial relationships between proposals to improve classification and location individually.

a) Visual relationship analysis: In large-scale object detection, visual reasoning is often used to solve the long tail distribution problem in datasets. A number of researches [60], [61], [62] utilized the visual features between proposals to construct a region-to-region reasoning graph for object detection, which benefitted a large number of categories in a single image. In order to perform visual reasoning in remote sensing images more effectively, we define the corresponding reasoning classes to the original classes. In this way, visual reasoning can be achieved by constructing the visual relationship between positive proposals, reasoning proposals, and negative proposals. Figs. 4 and 5 showed our visual reasoning ideas.

As shown in Fig. 4(a), the plane is difficult to detect due to its ambiguity; Fig. 4(b) shows the visual relationship between the plane positive proposal and its reasoning proposals. We can use this visual relationship for reasoning detection to improve the classification accuracy of the ambiguous plane. The positive proposal contains the major feature of the plane, and the reasoning proposals contain the local feature (wing and tail) or the global feature (airport). Through this visual feature, we can more easily reason that the category of the positive proposal is plane. In Fig. 5(a), the basketball court is small, which is easy to be missed or misdirected; Fig. 5(b) shows the visual relationship between the basketball court positive proposal and its reasoning proposals. We can use this visual relationship for reasoning detection to improve the classification accuracy of a small basketball court. The reasoning proposals contain the local feature or global feature of a basketball court, and other courts such as tennis courts and baseball courts which often appear nearby. This visual information can help us reason that the category of the positive proposal is more likely to be a basketball court. Therefore, it can be found that in remote sensing images, the visual relationship between proposals can be constructed for visual reasoning, so as to improve the classification accuracy.

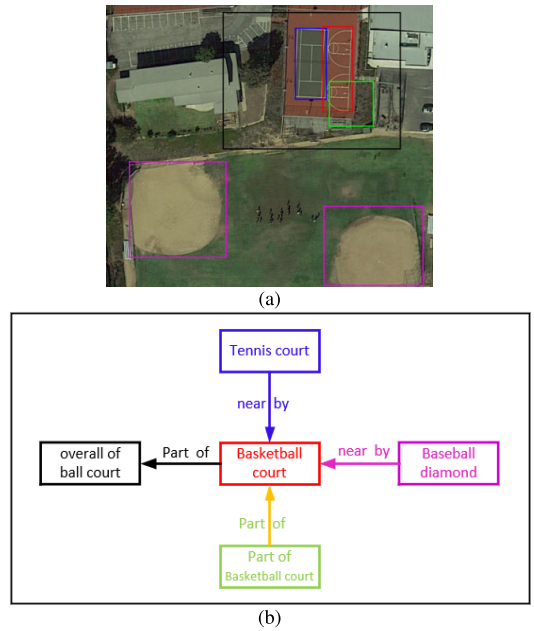
b) Spatial relationship analysis: The ambiguous and occluded objects in remote sensing images also make it difficult to achieve accurate localization, as low-quality objects cannot provide rich spatial information. In addition, the large-size objects in remote sensing images are too large to be located. As shown in Fig. 6(a), for a 1024 × 1024 pixels remote sensing image, the roundabout occupies almost half of the image. The roundabout is so large that it is difficult to be located. In visual relationship analysis, we explore the relationship between the visual features of proposals.



(a)

(b)

Fig. 4. Ambiguous plane. (a) Different proposals. The red box indicates the plane positive proposal, the black box indicates the reasoning proposal that contains the global feature, and the other boxes represent reasoning proposals at different positions of the plane. (b) Visual relationship between the plane positive proposal and reasoning proposals.



(a)

(b)

Fig. 5. Small basketball court. (a) Different proposals. The red box is a basketball court positive proposal, the blue box is a tennis court, the purple box is a baseball field, the green box contains the local features of the basketball court, and the black box contains the global features of the basketball court. (b) Visual relationship between basketball court positive proposal and reasoning proposals.

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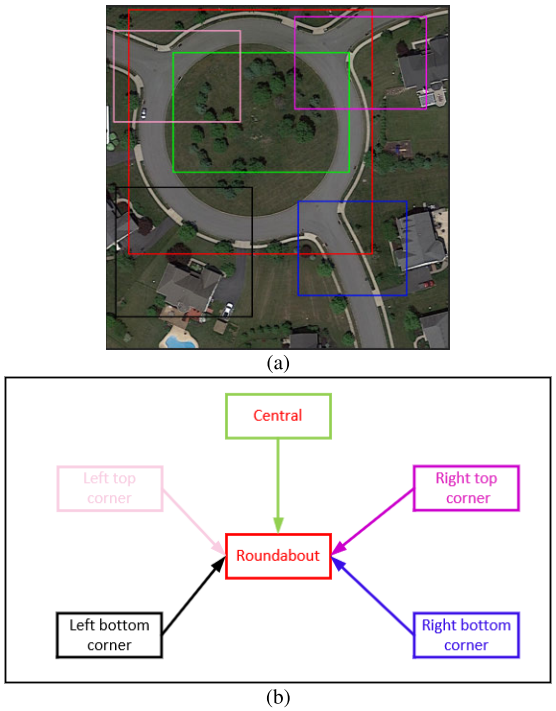


Fig. 6. Large roundabout. (a) Proposals in different positions of the roundabout. Among them, the red box is the positive proposal, the pink box is the top left corner of the roundabout, the purple box is the top right corner of the roundabout, the black box is the bottom left corner of the roundabout, the blue box is the bottom right corner of the roundabout, and the green box is the central position of the roundabout. (b) Spatial relationship between different position proposals.

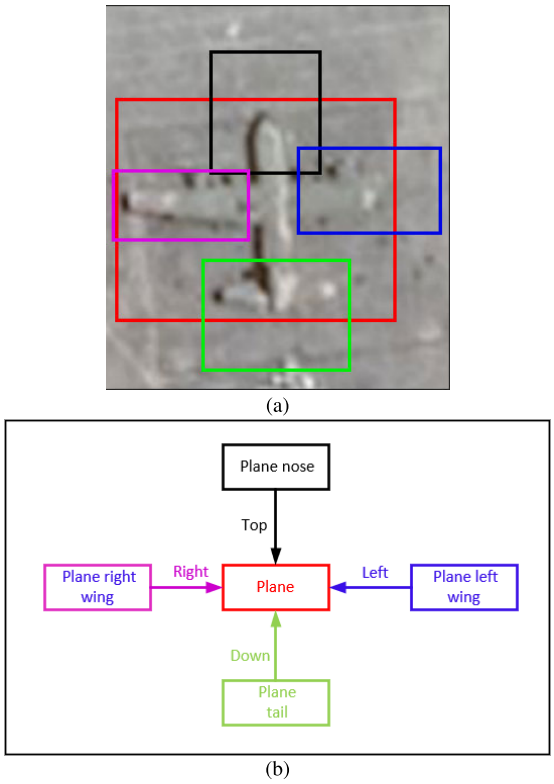


Fig. 7. Ambiguous plane. (a) Proposals at different positions of the plane, of which the red box is the plane positive proposal, the black box is the nose, the blue box is the left wing, the purple box is the right wing, and the green box is the tail. (b) Spatial relationship between different position proposals.

Similarly, there is a spatial relationship between proposals, which can be used to identify the position of the proposal. Fig. 6(b) is the spatial relationship between the roundabout proposals.

In Fig. 7(a), the plane is ambiguous and its color is similar to the ground, which makes it difficult to locate. Fig. 7(b) is the spatial relationship between the plane proposals. We can use the spatial relationship for reasoning object detection to improve the localization accuracy of ambiguous planes. For example, the right wing of the plane almost blends into the ground, so it is difficult to locate the plane. But we can infer the position of the plane from the position of its nose, tail, and left wing. Similar to visual relationship analysis, it is found that there is an obvious spatial relationship between proposals, and spatial reasoning can improve localization accuracy.

C. Graph Reasoning Module

In order to conduct visual reasoning and spatial reasoning simultaneously, we propose two GRMs to learn the visual reasoning features and spatial reasoning features individually. In the beginning, Faster RCNN is used for original classification and localization, and the weights of the original classifier and the locater are taken as the class-based visual semantic features and spatial semantic features. An overview of the GRM can be found in Fig. 8. In GRM, the visual relation and the spatial relation are formulated as two sparse region-to-region undirected graphs $G_{\text{cls}} = \{V_{\text{cls}}, A_{\text{cls}}\}$ and $G_{\text{loc}} = \{V_{\text{loc}}, A_{\text{loc}}\}$. The class-based semantic visual features and spatial features are mapped back to proposals by original classification score cls_score to get the node embeddings V_{cls} and V_{loc}, respectively. Two graph autoencoders (GAEs) are utilized to learn two interpretable sparse adjacency matrices which only retain the most relevant connections for object detection. Then, the visual and spatial embeddings of different regions are evolved and propagated by GCN, and the output of GRM is a reasoning feature.

1) Embedding of Proposals: Most graph-based approaches [63], [64], [65] propagated embeddings locally between proposals. However, it will fail if poor region embeddings lead to inefficient or wrong propagation. This situation often happens in object detection of remote sensing images when there are heavy occlusions and ambiguities in images. In order to alleviate this problem, we refer to the strategy of zero/few-shot learning [66], [67], [68], which utilizes the classifier's weights as the embedding or representation of an unseen/unfamiliar category. Unlike some existing works which only propagate the visual features on the image directly, we evolve the high-level semantic representations of all categories for both visual reasoning and spatial reasoning, so that to enhance the performances of both classification and localization. Specifically, we utilize the classifier's weights as the representation of categories.

a) Visual reasoning: Let $W_{\text{cls}} \in \mathbb{R}^{(2\times c+1)\times m_{\text{cls}}}$ denotes the visual features of the original classifier where c is the number of classes and m_{cls} is the dimension of the visual

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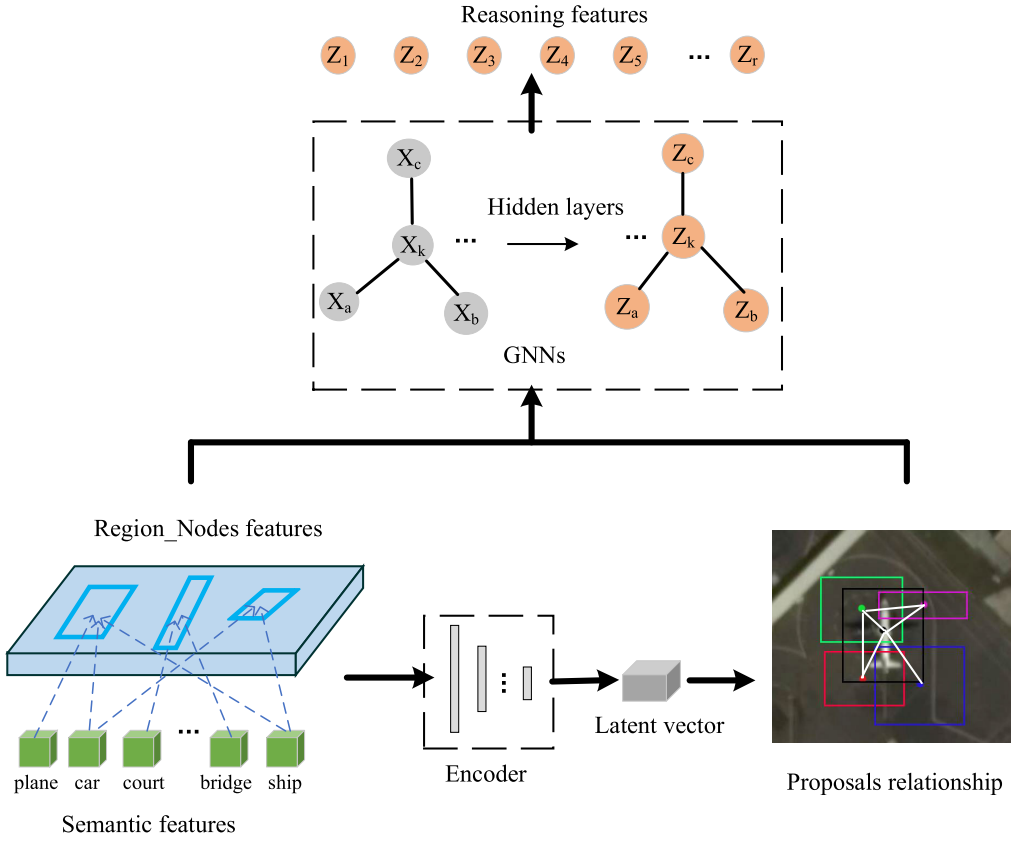


Fig. 8. Overview of the GRM. First, the class-based semantic visual or spatial features are mapped back to proposals. Then, the GAE is used to adaptively learn the relationship between proposals. Finally, the GCN is used to achieve the propagation of embeddings and obtain the reasoning features.

features. Since we construct corresponding reasoning classes for each class, there are $(2 \times c) + 1$ classes including background. Let G_{cls} be a region-to-region graph, we need to find an appropriate mapping from the class-based visual embeddings $w_{\text{cls}} \in W_{\text{cls}}$ to the regional embeddings of nodes $v^i_{\text{cls}} \in V_{\text{cls}}$. The softmax is used to convert the value of cls_score $\in \mathbb{R}^{r\times(2\times c+1)}$ (r is the number of proposals) between 0 and 1, then class-based visual embeddings are mapped into the proposals by the following equation:

$$V_{\text{cls}} = \text{softmax}(\text{cls_score})W_{\text{cls}}. \tag{1}$$

By softmax, cls_score can be regarded as the probability p_{cls} denoting the degree of proposals belonging to each class, then W_{cls} is mapped to each proposal through p_{cls} so that to obtain the visual embeddings V_{cls}.

b) Spatial reasoning: Most researches [63], [64], [65] constructed spatial embeddings of the proposals manually. For example, Li et al. [63] calculated the distance and the two centers' angle of the proposals as spatial embeddings. However, constructing embeddings manually is time-consuming. In this section, we use the original spatial features $W_{\text{loc}} \in \mathbb{R}^{(2\times c+1)\times k\times m_{\text{loc}}}$ to construct the spatial node embeddings, where k is the number of parameters to be regressed, and m_{loc} is the dimension of the proposals' features. In order to obtain the spatial semantic embeddings of each category, we first transform W_{loc} into $W_{\text{loc_reshape}} \in \mathbb{R}^{(2\times c+1)\times(k\times m_{\text{loc}})}$. Then, the reshaped $W_{\text{loc_reshape}}$ is mapped to each proposal by cls_score according to the following equation:

$$V_{\text{loc}} = \text{softmax}(\text{cls}_{\text{score}})W_{\text{loc_reshape}}. \tag{2}$$

Thus, we obtain the visual embeddings V_{cls} and the spatial embeddings V_{loc}, respectively.

2) Edges of Embeddings: To construct the G_{cls} and G_{loc}, we not only need the visual and spatial embeddings of each proposal, also need to obtain the adjacency matrices $A_{\text{cls}} \in \mathbb{R}^{r\times r}$ and $A_{\text{loc}} \in \mathbb{R}^{r\times r}$ for proposals. For simplicity, the regional embeddings and the adjacency matrices for the two graphs are uniformly defined as V and A. The adjacency matrices are adaptively learned by GAE [69], which is discussed in detail in this section.

First, we encode the regional embeddings to a latent space $Z \in \mathbb{R}^{d\times d}$ by GAE. Then, we use matrix multiplication as a decoder to learn the relation information ε between nodes by the following equation:

$$\varepsilon = ZZ^T, \quad \text{with} \quad Z = \text{encoder}(V) \tag{3}$$

where encoder is a nonlinear function. In this article, we consider two fully-connected layers with ReLU activation as the encoder. The latent vector Z is decoded into ε by matrix multiplication as $\varepsilon = ZZ^T$ so that the value of ε can be regarded as the similarity between regional embeddings.

Note that there are a lot of negative proposals in remote sensing images. ε is a fully connected adjacency matrix that also contains relationships between negative proposals. These redundant connections will reduce the efficiency of GRM.

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To solve this problem, we impose sparsity constraint on the graph, and let the first k largest values in each column of ε be 1 and the remaining values be 0

$$A_{i,j} = \begin{cases} 1, & \text{Top}_{k_{i=0,1,\cdot,r-1}}(e_{i,j}) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where $i, j \in [0, r-1]$. The constraint can effectively avoid redundant edge connections in the reasoning graph and ensure the rationality of relational reasoning.

3) Relational Reasoning: In order to obtain the reasoning features for object classification and localization, it is necessary to propagate visual and spatial embeddings globally over all the nodes in graphs. In this section, GCN is used for information propagation due to its strong feature extraction capability on graphs. GCNs are categorized into spectral approaches and spatial approaches. Spectral GCNs develop a graph convolution based on spectral graph theory. These approaches are inherently transduction and all graphs are required to share the same adjacency matrix. Spatial GCNs are inductive methods that can be adapted to different graph structures. Since different images generate different graph structures, spatial GCNs are more suitable for information propagate in our relational reasoning problem. Xu et al. [70] showed that GCNs were at most as powerful as the WL test [71] in distinguishing graph structures, and developed a graph isomorphism network (GIN) whose discriminative representational power was equal to the power of the WL test. Therefore, we use GIN for graph information propagation.

There are three ways of neighborhood embedding aggregation in GIN: sum, mean, and max. It was proved that sum aggregators can maximize the representation ability of the network. Therefore, we use sum as the aggregation scheme

$$h_i^{(k)} = \text{MLP}^{(k)}\left(h_i^{(k-1)} + \sum_{j \in \mathcal{N}(i)} h_j^{(k-1)}\right) \tag{5}$$

where $h_i^{(k)}$ is the feature vector of node i at the kth iteration layer and $\mathcal{N}(i)$ is a set of nodes adjacent to node i. Then, node features are embedded from low-level to high-level by multilayer perceptron (MLP). A two-layer GIN is designed for information propagation. After propagating, the visual reasoning feature $f'_{\text{cls}} \in \mathbb{R}^{r \times d_{\text{cls}}}$ and spatial reasoning feature $f'_{\text{loc}} \in \mathbb{R}^{r \times d_{\text{loc}}}$ for all the proposals can be obtained, where d_{cls} and d_{loc} is the dimensions of reasoning features f'_{cls} and f'_{loc} for each proposal.

Finally, f'_{cls} and f'_{loc} are concatenated to the proposal's original feature and the new features are fed into MLP to improve the classification and localization performance of object detection in remote sensing images.

IV. Experiments

This section provides a concise description of the experimental details, such as datasets, evaluation metrics, implementation details, and experimental results on the public remote sensing images datasets.

TABLE I
Categories and Indexes in DOTA and DIOR

DOTA		DIOR	
Category	Index	Category	Index
Plane	$C_1 1$	Airplane	$C_2 1$
Baseball diamond	$C_1 2$	Airport	$C_2 2$
Bridge	$C_1 3$	Baseball field	$C_2 3$
Ground track field	$C_1 4$	Basketball court	$C_2 4$
Small vehicle	$C_1 5$	Bridge	$C_2 5$
Large vehicle	$C_1 6$	Chimney	$C_2 6$
Ship	$C_1 7$	Dam	$C_2 7$
Tennis court	$C_1 8$	Expressway service area	$C_2 8$
Basketball court	$C_1 9$	Expressway toll station	$C_2 9$
Storage tank	$C_1 10$	Golf course	$C_2 10$
Soccer-ball field	$C_1 11$	Ground track field	$C_2 11$
Roundabout	$C_1 12$	Harbor	$C_2 12$
Harbor	$C_1 13$	Overpass	$C_2 13$
Swimming pool	$C_1 14$	Ship	$C_2 14$
Helicopter	$C_1 15$	Stadium	$C_2 15$
		Storage tank	$C_2 16$
		Tennis court	$C_2 17$
		Train station	$C_2 18$
		Vehicle	$C_2 19$
		Wind mill	$C_2 20$

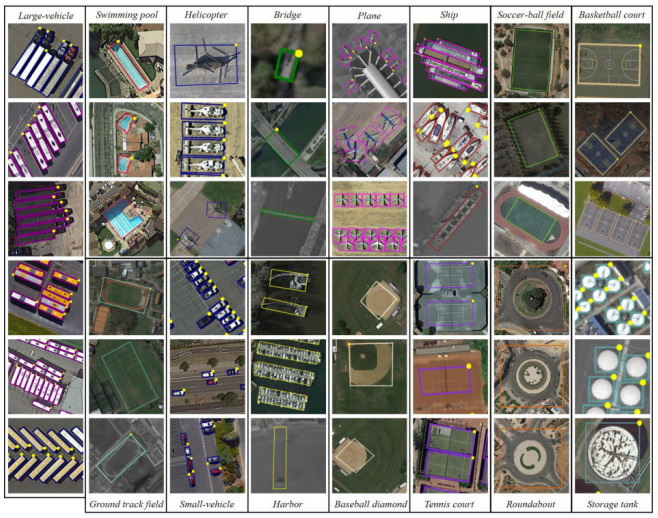
A. Datasets and Evaluation

Experiments were conducted on two large-scale object detection datasets of remote sensing images: DOTA [72] and DIOR [73]. DOTA contains 2806 large-size images from different sensors and platforms (e.g., Google Earth), and the size of images ranges from 800×800 pixels to 4000×4000 pixels. Object categories and their indexes are listed in Table I, from plane ($C_1 1$) to helicopter ($C_1 15$). The dataset is divided into three subsets for evaluation: half of the original images are chosen randomly as the training set, 1/6 of the dataset is as the validation set, and the remaining 1/3 images are composed of the testing set. To reduce the computational complexity, original images are cut into 1024×1024 pixels in the experiments, and the evaluation results are obtained by the DOTA evaluation server. Fig. 9(a) shows the annotated samples of different categories in DOTA.

DIOR is also a large-scale remote sensing dataset for object detection. The size of images is 800×800 pixels, object categories and their index names are listed in Table I, from an airplane ($C_2 1$) to wind mill ($C_2 20$). The DIOR contains 23 463 annotated images and 190 288 instances. The training set is composed of randomly selected 1/4 original images, 1/4 original images are randomly selected as the validation set, and the remaining 1/2 as the testing set. Fig. 9(b) shows the annotated samples of different categories in DIOR.

The bounding boxes can be divided into four classes: true positives (TPs), false positives (FPs), true negatives (TNs), and

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(a)



(b)

Fig. 9. Samples of annotated images in (a) DOTA and (b) DIOR.

false negatives (FNs). A positive bounding box is considered a TP when the IoU between the predicted bounding box and the ground truth is greater than the threshold. Otherwise, it is regarded as an FP. A negative bounding box is considered a TN when the IoU between the predicted bounding box and the ground truth is less than the threshold. Otherwise, it is regarded as a FN. Precision and recall are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{(\text{TP} + \text{FP})} \tag{6}$$

$$\text{Recall} = \frac{\text{TP}}{(\text{TP} + \text{FN})}. \tag{7}$$

The horizontal axis of the P–R curve is recall, the vertical axis is precision, and the average precision (AP) is the area under the P–R curve. Normally, recall and precision are negatively correlated. AP is used to quantitatively evaluate the performance of the object detection model for each class. Mean AP (mAP) evaluates the overall performance of all object classes. It is calculated by averaging all categories' AP, i.e.,

$$\text{mAP} = \frac{1}{n}\sum_{i=1}^{n} \text{AP}_i \tag{8}$$

where n is the number of classes and AP_i is the AP of class i.

B. Implementation Details and Parameter Optimization

Our proposed relational reasoning strategy can be stacked on any two-stage or one-stage deep learning-based object detection model without any prior knowledge. In this section, we selected several benchmarks to verify the effectiveness of DGRN, including Faster RCNN [8], Cascade RCNN [12], RetinaNet [31], YOLO v3 [14], ROI-Trans [5], YOLOX [35], SASM [74], YOLO v7 [75], and DETR [38]. All the experiments were conducted on NVIDIA Titan RTX cards using pytorch [76]. One-stage model (such as RetinaNet, YOLO v3, YOLOX, and YOLO v7) divide the images into proposals and predict bounding boxes and their probabilities simultaneously. Two-stage models (such as Faster RCNN, Cascade RCNN, and ROI-Trans) first transform anchors into proposals through the RPN and then predict bounding boxes and probabilities for each proposal. The transformer-based object detection model, such as DETR, combines convolutional neural networks and transformer encoders–decoders. To generate reasoning proposals that are needed in DGRN, we modified the general sampling method. If the IoU of a proposal is greater than 0.5, it is considered as a positive proposal. If the IoU of a proposal is less than 0.2, it is considered as a negative proposal. Otherwise, it is considered as a reasoning proposal. In our experiments, a two-stage model sampled about 512 ROIs, and ROI Align was used to extract ROIs' features. Features of each proposal detected by a one-stage model were extracted directly from the feature map. Then, the features of the ROIs or proposals were fed into two full connection layers for object classification and regression. Different from horizontal object detection, the rotated object detection model could not only predict the coordinates of the center points, length, and width of the object but also the object angle. The mAP is utilized as an evaluation matrix for both horizontal object detection and rotated object detection.

The settings of the GRM were consistent in all the compared benchmarks. The GRM took the weights of the classifier and locater of the benchmark as visual embeddings and spatial embeddings. GAE was applied to learn the relationship between nodes, and GIN was conducted to achieve relational reasoning for object detection.

C. Experimental Results and Analysis

1) Comparison With Benchmarks: In order to verify the effectiveness of DGRN for object detection in remote sensing images, we compared it with nine benchmark models on DOTA and DIOR, including Faster RCNN, Cascade RCNN, RetinaNet, YOLO v3, YOLOX, YOLO v7, ROI-Trans, DETR, and SASM. Tables II and III showed the results of various models on DOTA and DIOR. "Baseline" was the benchmark, "DGRN" meant implementing the idea of our method based on the benchmark.

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TABLE II
COMPARISON OF AP (%) FOR VARIOUS OBJECT DETECTION MODEL ON DOTA

Category	Faster RCNN		Cascade RCNN		RetinaNet		YOLO v3		ROI-Trans		SASM	
	Baseline	DGRN	Baseline	DGRN	Baseline	DGRN	Baseline	DGRN	Baseline	DGRN	Baseline	DGRN
$C_1 1$	88.1	**89.3**	88.3	**89.9**	87.1	**88.8**	83.6	**84.9**	**88.9**	88.2	90.5	**90.9**
$C_1 2$	77.0	76.0	77.9	**78.4**	78.0	**80.0**	**72.6**	71.2	84.5	**85.1**	85.8	**86.4**
$C_1 3$	**46.4**	45.4	**52.6**	52.1	**44.0**	43.2	41.3	**42.9**	54.2	**55.5**	**70.3**	69.8
$C_1 4$	62.0	**69.9**	65.4	**71.1**	65.3	**67.1**	68.3	**71.8**	76.5	**77.4**	84.1	**84.3**
$C_1 5$	76.4	**77.5**	77.2	**78.3**	73.9	**74.5**	75.2	**76.3**	78.5	**78.8**	72.4	**72.9**
$C_1 6$	71.1	70.6	**76.4**	76.2	59.6	**61.0**	**58.1**	57.3	77.6	**78.2**	61.9	**63.4**
$C_1 7$	84.1	84.4	91.0	**91.3**	79.9	**82.3**	**81.9**	80.3	**87.8**	86.8	**74.3**	74.2
$C_1 8$	90.8	**90.9**	90.1	**90.4**	89.6	**90.1**	85.7	**86.1**	**90.8**	89.5	89.2	**89.7**
$C_1 9$	76.8	**81.3**	72.4	**80.3**	68.5	**75.3**	70.2	**71.5**	86.7	**87.1**	82.3	**82.5**
$C_1 10$	80.8	**84.8**	79.6	**82.8**	64.9	**70.8**	64.0	**64.8**	85.6	**86.3**	78.8	**79.4**
$C_1 11$	39.9	**44.1**	37.7	**43.3**	43.5	**47.8**	43.1	**45.7**	62.5	**64.7**	63.0	**63.9**
$C_1 12$	59.7	**61.6**	63.0	**64.5**	61.9	**63.2**	64.8	**66.3**	61.4	**63.2**	82.1	**82.5**
$C_1 13$	63.4	**64.5**	65.3	**66.0**	**60.7**	59.9	49.2	**50.3**	75.2	**76.0**	**67.9**	67.4
$C_1 14$	56.1	**66.3**	56.0	**65.0**	53.2	**55.8**	56.2	**57.8**	70.9	**71.4**	**83.3**	83.0
$C_1 15$	44.6	**54.0**	47.2	**56.1**	32.1	**43.2**	28.1	**31.1**	54.1	**54.8**	65.4	**65.9**
mAP	67.8	**70.7**	69.3	**72.3**	64.1	**66.8**	61.9	**63.9**	75.7	**76.2**	76.8	**77.8**

a) Detection accuracy: The DGRN proposed was applicable to both the horizontal and the rotated box object detection task. The results in Tables II and III showed that our DGRN method outperformed the benchmark model by 1%–3% on DOTA and DIOR. On DOTA, our method achieved an overall AP of 70.7% compared to 67.8% by Faster RCNN, 72.3% compared to 69.3% by Cascade RCNN, 66.8% compared to 64.1% by RetinaNet, 63.9% compared to 61.9% by YOLO v3, 76.2% compared to 75.7% by ROI-Trans, and 77.8% compared to 76.8% by SASM. On DIOR, our method achieved an overall AP of 73.9% compared to 71.5% by Faster RCNN, 75.4% compared to 73.2% by Cascade RCNN, 69.2% compared to 67.6% by RetinaNet, 62.2% compared to 60.8% by YOLO v3, 67.6% compared to 66.7% by YOLOX, 71.8% compared to 71.3% by DETR, and 69.7% compared to 69.0% by YOLO v7. It demonstrated that DGRN could be stacked on a deep learning-based object detection model conveniently, and improved the performance of the basic model.

In addition, we compare the improved benchmark model based on DGRN with other object detection methods in recent years, including rotated box object detection and horizontal box object detection. In the DOTA dataset, we integrated the benchmark method (DGRN) with the rotated box object detection model, such as S2ANet [55], R3Det [54], Oriented RCNN [57], ReDet [56], Oriented RepPoints [59], and SASM [74] for comparison. S2A-Net designed two modules: a FAM (generating high-quality anchors) and an ODM (encoding the orientation information). R3Det used a progressive regression approach from coarse to fine granularity and designed a feature refinement module to improve detection performance by getting more accurate features. Oriented R-CNN was a general two-stage oriented detector with promising accuracy and efficiency. ReDet addressed the highly redundant and inefficient issues, which explicitly encoded rotation equivariance and rotation invariance. Oriented RepPoints used an effective adaptive points learning approach to aerial object detection by taking advantage of the adaptive points representation. SASM utilized new flexible shape-adaptive selection (SA-S) and shape-adaptive measurement (SA-M) strategies for object-oriented detection.

In the DIOR dataset, we integrated the benchmark method (DGRN) with the horizontal box object detection model, such as CANet [77], MFPNet [78], CF2PN [79], PANet [80], T-TRD-DA [81], DETR [38], and YOLO v7 [75] for comparison. CANet obtained stronger features for better small object detection. MFPNet integrated both local and global features to detect scattered objects and targets with scale-differences in remote sensing images. CF2PN extracted sufficiently comprehensive semantic information for multiscale fusion. PANet was proposed to boost the entire feature hierarchy with accurate localization information in lower layers by bottom-up path augmentation, which can significantly shorten the information path between lower layers and topmost features. T-TRD-DA designed aggregate features of global spatial positions on multiple scales and modeled the interactions between pairwise instances. DETR proposed a new architecture for Transformer object detection based on encoder and decoder. The YOLO v7 model was motivated by YOLO v5, and it improves the accuracy of object detection by a more robust loss function, a more efficient label assignment strategy, and a more effective training technique. As shown in Fig. 10, SASM (DGRN) obtained the top mAP on the DOTA rotated box dataset and Cascade RCNN (DGRN) obtained the best mAP on the DIOR horizontal box dataset, demonstrating that DGRN

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TABLE III
COMPARISON OF AP (%) FOR VARIOUS OBJECT DETECTION MODEL ON DIOR

Category	Faster RCNN		Cascade RCNN		RetinaNet		YOLO v3		YOLOX		DETR		YOLO v7	
	Baseline	DGRN	Baseline	DGRN	Baseline	DGRN	Baseline	DGRN	Baseline	DGRN	Baseline	DGRN	Baseline	DGRN
C_21	65.9	**67.4**	67.1	**68.1**	64.2	**65.2**	62.3	**64.2**	64.4	**65.6**	63.9	**64.2**	67.2	**67.8**
C_22	**79.8**	78.8	79.2	**79.6**	**80.2**	79.1	**30.2**	29.5	**34.6**	33.5	**74.2**	73.8	**74.6**	74.0
C_23	72.7	**74.2**	73.8	**75.1**	74.1	**75.2**	68.1	**70.6**	71.4	**72.3**	76.1	**76.5**	73.1	**73.7**
C_24	88.2	**90.2**	86.9	**88.1**	79.5	**82.4**	74.5	**76.9**	80.2	**80.7**	80.1	**80.3**	85.4	**86.2**
C_25	48.5	**50.2**	52.3	**55.7**	46.3	**48.9**	39.1	**41.1**	48.6	**50.2**	48.0	**49.3**	46.1	**47.3**
C_26	78.8	**80.1**	81.8	**87.4**	66.0	**68.8**	62.9	**66.2**	64.6	**66.9**	77.1	**78.0**	76.6	**77.5**
C_27	64.1	**66.2**	64.0	**65.8**	62.1	**63.9**	45.7	**46.3**	45.6	**46.2**	65.9	**66.7**	62.7	**64.1**
C_28	**82.2**	81.1	83.1	**83.7**	**80.1**	79.8	**71.3**	70.2	**80.4**	79.5	**76.4**	75.8	79.2	**80.6**
C_29	70.5	**71.3**	**74.1**	74.0	**71.0**	69.8	65.2	**66.8**	72.4	**73.1**	75.1	**75.6**	73.2	**73.8**
C_210	78.3	**81.8**	78.2	**82.2**	69.2	**72.5**	50.3	**54.4**	78.5	**80.8**	79.2	**80.8**	78.2	**78.5**
C_211	82.5	**87.5**	85.8	**92.6**	84.2	**87.7**	83.4	**85.6**	**86.5**	86.4	**78.8**	77.5	**81.6**	80.4
C_212	60.9	**63.8**	61.6	**63.2**	57.9	**60.3**	55.3	**56.4**	58.4	**60.3**	63.8	**64.5**	50.7	**51.6**
C_213	62.5	**65.1**	64.1	**68.0**	63.3	**67.1**	58.2	**60.4**	61.4	**62.0**	62.1	**62.9**	59.5	**61.2**
C_214	76.0	**79.1**	74.0	**76.3**	**71.2**	70.4	**73.9**	72.0	73.2	**74.1**	74.9	**75.8**	73.3	**74.0**
C_215	77.2	**84.2**	78.1	**82.6**	78.1	**79.2**	74.1	**78.4**	74.3	**76.3**	74.1	**74.3**	63.4	**63.7**
C_216	63.5	**67.2**	62.9	**68.1**	43.9	**49.2**	44.2	**46.1**	55.3	**56.4**	64.2	**64.9**	58.5	**60.2**
C_217	88.0	**88.2**	88.4	**88.9**	**88.3**	88.1	88.2	88.2	**88.5**	88.2	**88.2**	87.7	85.9	**86.2**
C_218	61.7	**63.0**	64.4	**65.1**	51.3	**52.0**	41.0	**41.8**	62.5	**63.6**	68.0	**68.5**	61.9	**62.3**
C_219	44.2	**52.2**	56.8	**60.2**	39.8	**44.2**	43.9	**45.1**	46.1	**46.9**	55.7	**56.3**	42.9	**43.1**
C_220	86.3	**87.1**	**88.1**	86.1	**81.3**	79.3	**84.2**	83.9	88.7	**88.9**	80.5	**81.6**	86.9	**87.4**
mAP	71.5	**73.9**	73.2	**75.4**	67.6	**69.2**	60.8	**62.2**	66.7	**67.6**	71.3	**71.8**	69.0	**69.7**

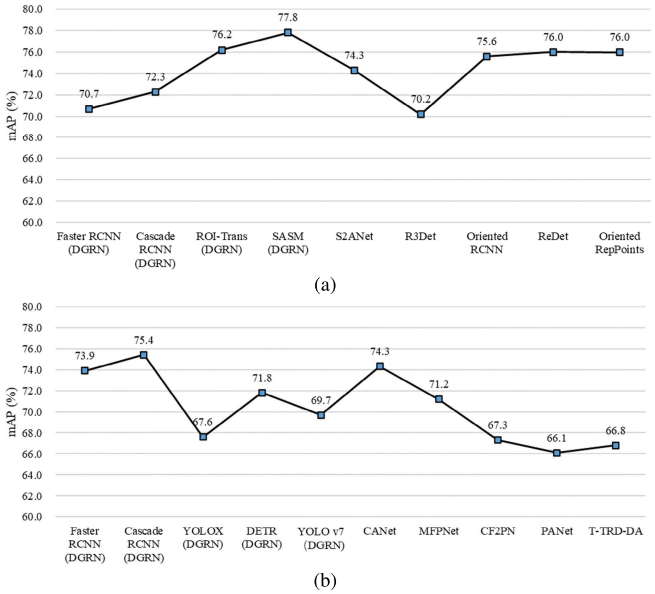


Fig. 10. Comparison of mAP (%) for other object detection models on (a) DOTA and (b) DIOR.

was very competitive with the existing model on different tasks.

In addition, we implemented the idea of SGRN [61] to examine the effect of general reasoning methods on remote sensing images. The comparison results were shown in Fig. 11, where "SGRN" meant implementing the idea of SGRN based on the benchmark. It can be seen that SGRN based on Faster RCNN can only improve 1.2% on DOTA and 0.2% on DIOR. Our DGRN enhanced the accuracy of the Faster RCNN model by 2.9% on DOTA and 2.4% on DIOR. The categories in a remote sensing image are insufficient, resulting in the performance of SGRN was not good enough. It showed that the reasoning proposals defined in DGRN played an important role in relational reasoning. The performance of DGRN on each category also illustrated that our DGRN method improved the object detection performance of most classes, but the accuracies were slightly decreased on a few classes, such as bridge. It is observed that the misclassified classes were too similar in background. For example, the AP of C_13 (bridge) got by DGRN was slightly decreased compared to the benchmark. Bridges could be seen as background for other classes such as vehicles, which made our method unable to detect them correctly. Fig. 12 showed qualitative results comparison between the Faster RCNN and Faster RCNN+DGRN, Cascade RCNN and Cascade RCNN+DGRN, RetinaNet and RetinaNet+DGRN, YOLOv3

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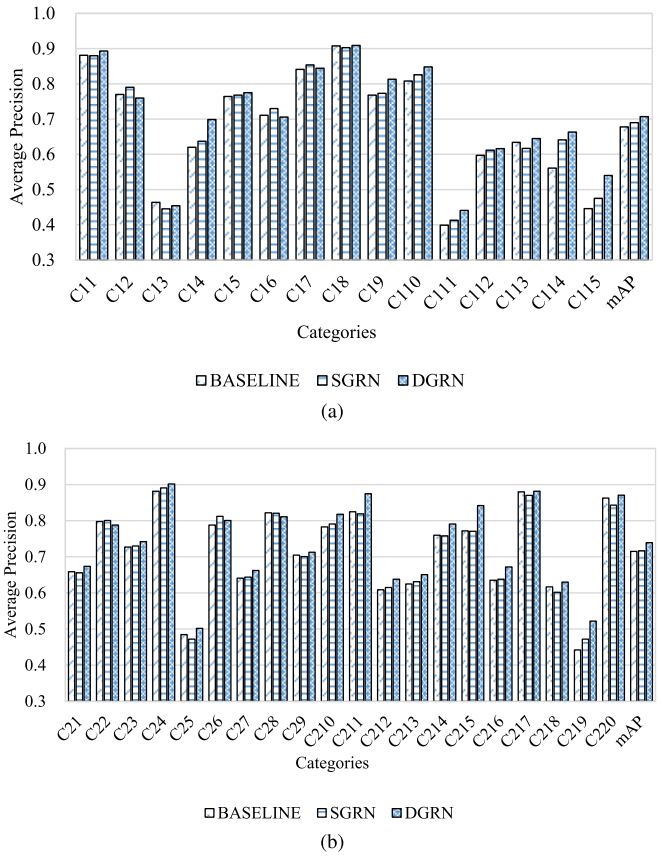


Fig. 11. Comparison of SGRN and DGRN of AP on (a) DOTA and (b) DIOR.

and YOLOv3+DGRN on DOTA. Proposals of different colors represented different categories. It could be found that our method has better detection results for most of the categories with obvious differences from the background. In addition, we found that our method improved the accuracy of small objects not as good as that of large objects.

b) Comparison between one-stage and two-stage model: From Tables II and III, it could be more intuitively found that our DGRN method had an obvious performance improvement on the two-stage benchmark model. The DGRN improved about 3% better than the two-stage benchmark model, while it improved about 2% better than the one-stage benchmark model. It may be due to the fact that the two-stage model generated proposals through the RPN, which eliminated a large number of negatives. This made the two-stage model establish more high-quality regions relational graph, and carry out reasoning detection more effectively. It can be seen from Fig. 12 that the results of RetinaNet were slightly worse than those of Faster RCNN. However, the accuracies of object detection of remote sensing images were enhanced obviously by both DGRN stacked on two-stage and one-stage benchmark models.

c) Computational complexity: There are two major modifications to our method from the benchmark model: detection of reasoning proposals and GRM. We compared the computational speed (training and testing), parameters, and FLOPS in Table IV. As could be seen in Table IV, those modifications did not excessively increase the computational complexity

TABLE IV
COMPARISON OF BENCHMARKS AND DGRN ON SPEED, PARAMETERS, AND FLOPS

Method		Train speed	Test speed	Parameters	FLOPS
Faster RCNN	Baseline	0.3s	0.13s	6762MB	83.13GB
	DGRN	0.33s	0.15s	10083MB	126.95GB
Cascade RCNN	Baseline	0.38s	0.17s	7817MB	95.99GB
	DGRN	0.41s	0.18s	12053MB	152.01GB
RetinaNet	Baseline	0.26s	0.08s	4146MB	81.66GB
	DGRN	0.28s	0.09s	7258MB	145.83GB
YOLO v3	Baseline	0.23s	0.07s	3998MB	61.95GB
	DGRN	0.24s	0.08s	6675MB	110.43GB

TABLE V
COMPARISON OF BENCHMARK MODEL AND DGRN ON AR (%)

Method		AR_{small}	AR_{medium}	AR_{large}
Faster RCNN	Baseline	**28.4**	53.2	76.6
	SGRN	28.3	52.2	75.3
	DGRN	27.1	**53.8**	**78.3**
Cascade RCNN	Baseline	**30.5**	54.3	78.6
	DGRN	29.8	**54.6**	**80.2**
RetinaNet	Baseline	**18.1**	**45.4**	71.9
	DGRN	17.2	45.2	**73.5**
YOLO v3	Baseline	**20.3**	**45.5**	70.4
	DGRN	19.8	44.3	**71.6**

compared to benchmarks. All the speeds were tested on images with the size of 1024×1024 pixels on a single TITAN RTX. For a single image, our method only increased 0.01–0.03 s of time and 3000 MB memory during training and testing. Therefore, our method was light-weighted and did not impose too much computational burden on the model.

d) Recall rate (AR): The AR measures the proportion of positives that are correctly detected. Since DOTA did not provide the annotation of the test set, we calculated the AR of the different models on the DIOR dataset, and the results were listed in Table V. AR_{small} was the AR of objects with an area smaller than 32^2, AR_{medium} was the AR of objects with an area between 32^2 and 96^2, and AR_{large} was the AR of objects with an area larger than 96^2. As shown in Table V, our method had a better AR for larger objects. For the two-stage model, our method improved the AR by about 2% for large objects and about 1% for medium objects. For the one-stage model, our method improved the AR of large objects by around 1.5%. However, our method could not effectively improve the AR of smaller objects and may have decreased slightly compared with the benchmark model. We considered that smaller objects contained less relational information, which made it difficult to carry out relational reasoning detection effectively.

2) Ablation Studies: In this section, we conducted a series of ablation experiments on DOTA to evaluate the effectiveness

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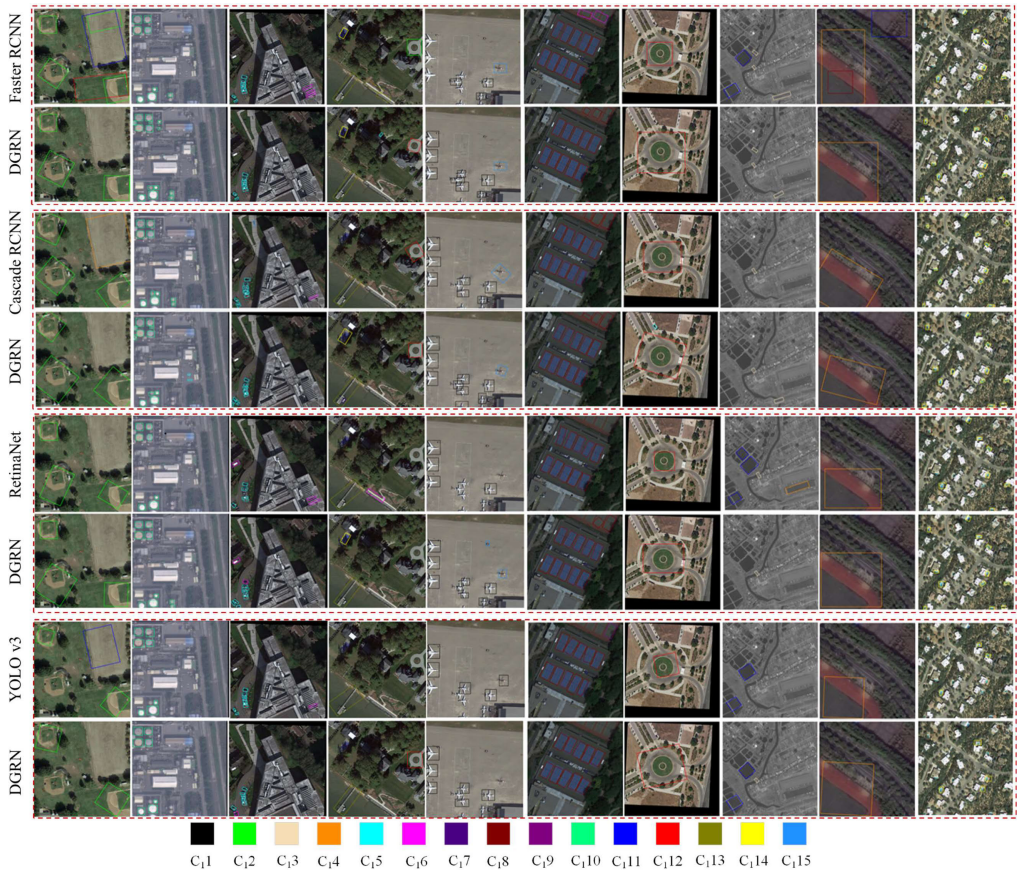


Fig. 12. Qualitative result comparison on DOTA.

TABLE VI
RESULTS OF ABLATION STUDIES

Optional settings	Baseline	Different settings				
RPD	Faster RCNN	✓	✓	✓		✓
VRR			✓		✓	✓
SRR				✓	✓	✓
mAP (%)	67.8	68.5	70.3	70.1	69.1	70.7

of various modules. The Faster RCNN with FPN was considered as the baseline for the experiments. Table VI showed the results of ablation studies. RPD indicated reasoning proposals detection, VRR indicated visual relationship reasoning, and SRR indicated spatial relationship reasoning.

a) Reasoning proposals detection: Detection of reasoning proposals slightly improved the mAP by 0.7%. This was because reasoning proposals alleviated the imbalance of positive and negative samples in remote sensing images, and detecting more "hard examples" (reasoning proposals) could improve the robustness of the model. In addition, relational reasoning with detecting reasoning proposals achieved 70.7% mAP compared to 69.1% without detecting reasoning proposals. This proved that reasoning proposals played a vital role in relational reasoning.

b) Relational reasoning: Our method carried out visual relational reasoning and spatial relational reasoning to improve classification and localization, respectively. As shown in Table VI, with detecting reasoning proposals, visual reasoning could significantly improve the mAP by 2.5%, spatial reasoning could significantly improve the mAP by 2.3%, and both of them can improve the mAP by 2.9%. This showed that both visual features and spatial features could enhance the performance of our model in relational reasoning detection. However, we also found that the performance of the model was not greatly improved by only conducting visual reasoning and spatial reasoning simultaneously, without reasoning proposals detection.

c) Sparsity constraints: Due to the large number of redundant background boxes in remote sensing images, it is not possible to connect all the candidate box nodes. These redundant connections will reduce the efficiency of the GRM. We address this issue by sampling the similarity matrix between nodes. This section explored the impact of different numbers of sampling. In this section, we performed experiments using $1/N$ of the total number of ROIs (6144), where N is set to 4, 6, 8, and 10, respectively, and the experimental results were shown in Table VII. The experimental results showed that when N is 8, the redundant edge connections in the relational graph can be effectively avoided, ensuring rational and efficient relational reasoning.

d) Visualization analysis of DGRN: The heatmaps are shown in Fig. 13 to further illustrate the effectiveness of DGRN. The first column demonstrates the original images, and

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TABLE VII
EVALUATION OF SPARSITY CONSTRAINT

N	Number of ROIs	TopK	mAP (%)
4	6144	1536	70.5
6		1024	69.8
8		768	70.7
10		614	68.1

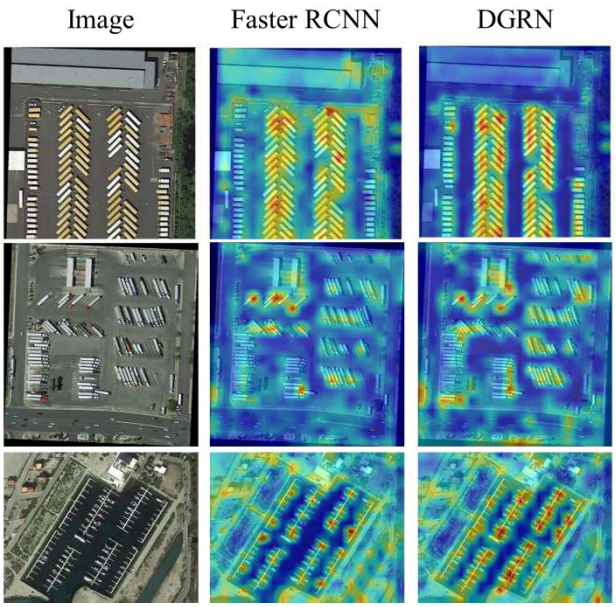


Fig. 13. Heat maps of Faster RCNN and Faster RCNN + DGRN on DATA.

the second column and third column represent the heatmaps of Faster RCNN and Faster RCNN+DGRN, respectively. It could be seen that the cars in the parking lot are densely distributed in remote sensing images, and thus the semantic information in the heatmaps of Faster RCNN are ambiguous around their edges. We could observe that the central features of the cars are enhanced, and the ambiguous features of the edges are suppressed by introducing DGRN. As a result, the boundaries between adjacent objects become clearer, which contributes to the bounding box to accurately cover the densely distributed objects. DGRN highlights features around the center of the indistinctive objects, which enables the network to focus on acquiring more effective information.

The visualization performances of object detection results in Fig. 12 also demonstrate the advantage of our proposed DGRN. In the cases of the baseline model the objects are easily disturbed by the background, the DGRN-enhanced model could obtain more accurate object detection performances. DGRN is guided by the GNN mechanism to capture characteristic features by exploring the relationship between the center of objects and their backgrounds so as to obtain high discrimination capability for object detection in remote sensing images.

V. DISCUSSION AND OUTLOOK

In this article, we introduce a novel relational reasoning network for object detection in remote sensing images, i.e., DGRN. The DGRN considers both visual relationships and spatial relationships between proposals simultaneously, by introducing reasoning proposals and a GRM. Reasoning proposals detection gives the DGRN the ability to fully capture the information of the foreground, and the GRM uses a graph convolution neural network to propagate embeddings between all proposals in the images. The DGRN can be added to a two-stage or one-stage deep learning-based object detection model without prior knowledge, to give a benchmark the capability of reasoning detection. Plenty of experiments were conducted on DOTA and DIOR, and the results suggested that reasoning detection is helpful in enhancing the object detection performance of remote sensing images. The DGRN could be considered as a reasoning plug-in, which is flexible and effective. Compared to baseline networks, introducing the DGRN module will be slightly more computationally intensive due to the graph convolutions. However, it is lightweight and does not impose too much computational burden generally.

Although DGRN has been successfully applied to several popular object detection models in remote sensing images, such as Faster RCNN, Cascade RCNN, RetinaNet, YOLO, and ROI-Trans, there's no guarantee that DGRN could enhance the performances of all deep learning-based detection model. The generalization ability of DGRN could be improved in the future. Furthermore, we will continue to deepen the mining of visual and spatial relationships to make the model more interpretable, and we will pay more attention to effective feature representation to make reasoning detection more comprehensive.

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